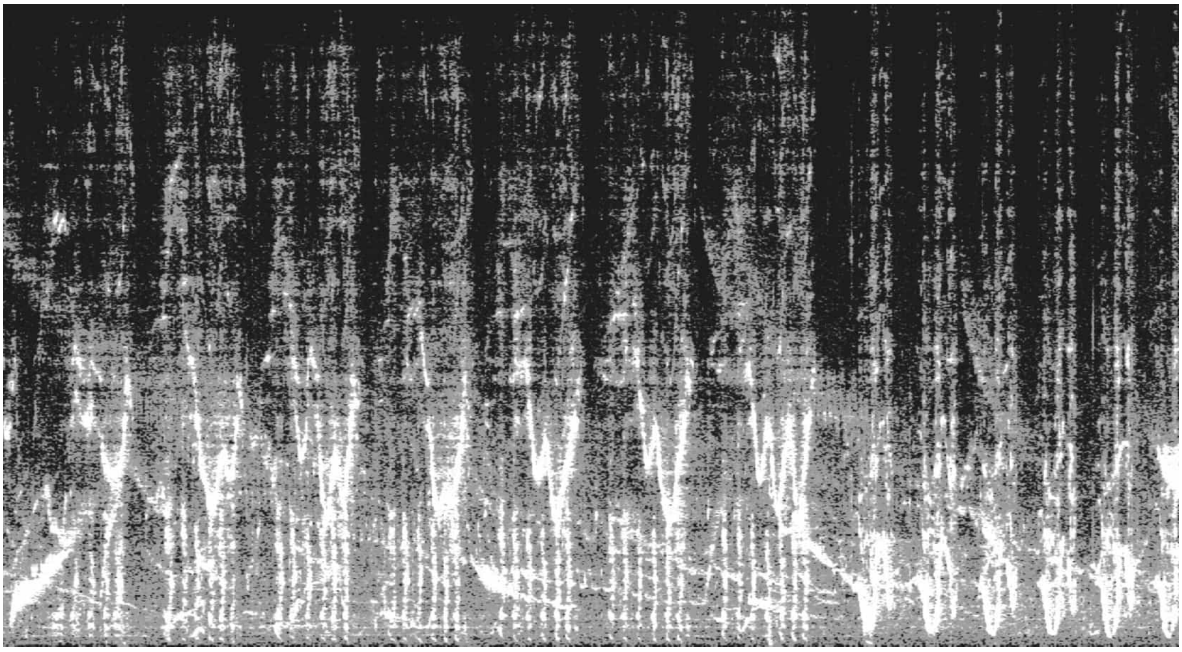


Credit 1.47

47. A spectrogram of a bowhead whale song. How does a whale experience this song? The *Voyager* record included a whale song in addition to Beethoven, Bach and Chuck Berry. We can only hope it is a good one.

None of this should surprise us. Sapiens don't rule the world because they have deeper emotions or more complex musical experiences than other animals. So we may be inferior to whales, bats, tigers and pelicans at least in some emotional and experiential domains.

Beyond the mental spectrum of humans, bats, whales and all other animals, even vaster and stranger continents may lie in wait. In all probability, there is an infinite variety of mental states that no Sapiens, bat or dinosaur ever experienced in 4 billion years of terrestrial evolution, because they did not have the necessary faculties. In the future, however, powerful drugs, genetic engineering, electronic helmets and direct brain–computer interfaces may open passages to these places. Just as Columbus and Magellan sailed beyond the horizon to explore new islands and unknown continents, so we may one day set sail towards the antipodes of the mind.